ENGINEERING PORTFOLIO

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I am an electrical engineer who enjoys problems that invite different perspectives. The projects I have chosen throughout my undergraduate studies have led me to signal processing, embedded systems, machine learning, networking, communications and audio.

Since most of my projects center on manipulating and filtering signals, I am seeking an internship connected to that work. This semester, I am focused on the EE 31 Junior Design project, in which I am building an autonomous robot, and on a MIDI instrument for EE 95, which lets me explore how technology can be both creative and unconventional.



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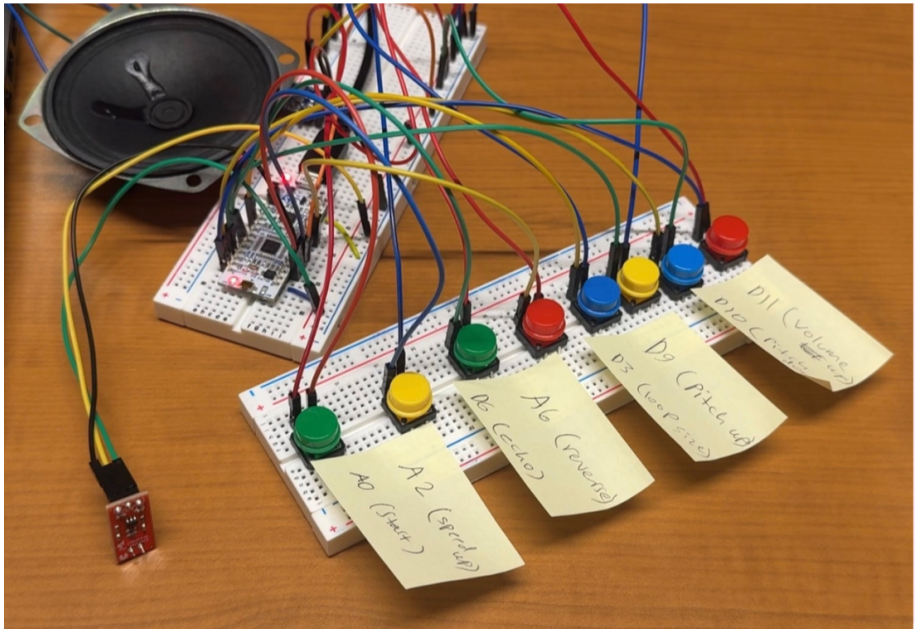
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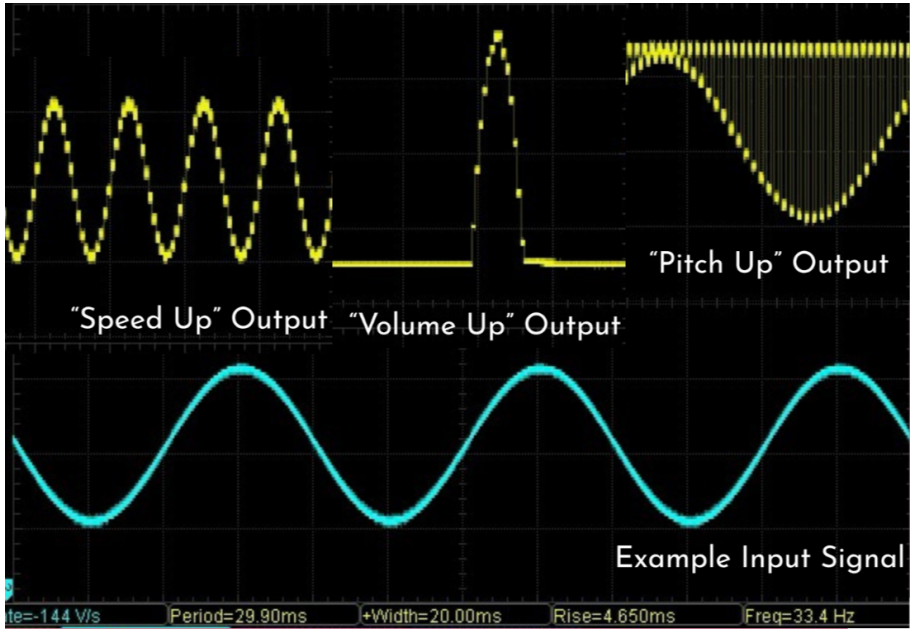
PROJECT 1 | EE 14 EMBEDDED SYSTEMS | JAN. 2026 TO MAY 2026

DJ Board on an STM32

A real-time audio effects board that records sound and plays it back with effects like echo, reverse, pitch and speed, all within the STM32's 64KB of on-chip memory. Playback first came out with a high-pitched hiss and clicks, so I wrote a timer-driven state machine that records and plays every sample at a steady rate, and faded each clip in and out. **An oscilloscope confirmed that the playback matched the original recording.**



The board: microphone, speaker and effect buttons

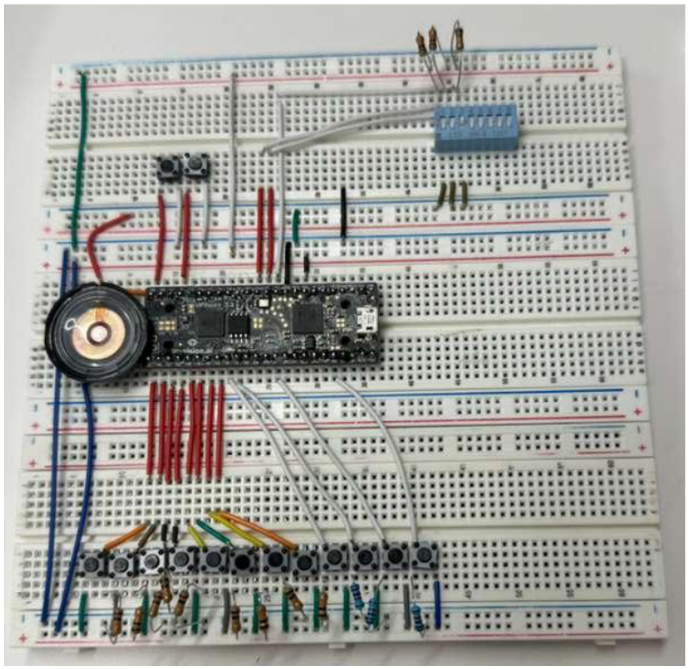


Oscilloscope captures of the effect outputs

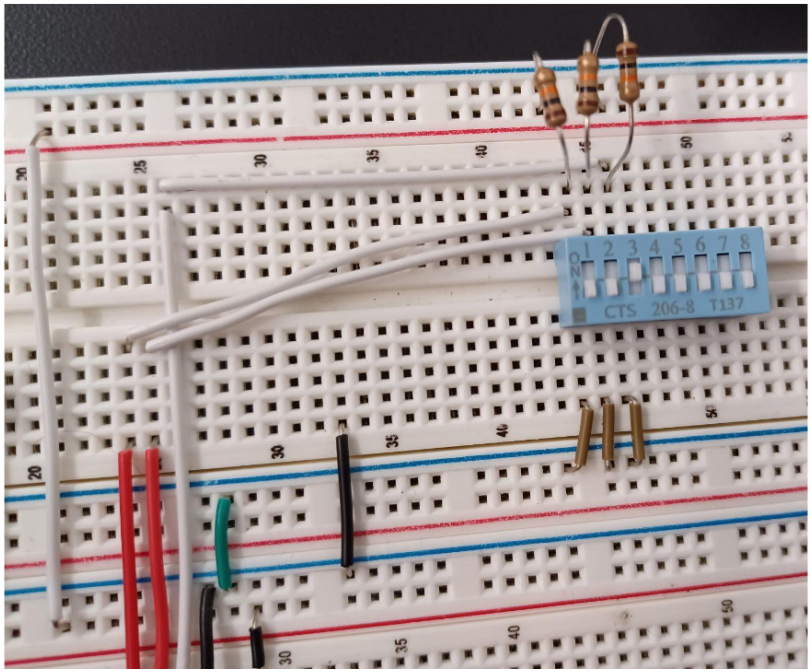
PROJECT 2 | ES 4 DIGITAL LOGIC CIRCUITS | SEPT. 2025 TO DEC. 2025

FPGA Audio Synthesizer

An 8-octave synthesizer, from 16Hz to 8kHz, written in SystemVerilog for a MIDI keyboard. My first design divided on the chip to calculate every note and needed over 250% of the FPGA's space. Since dividing by 2 is the same as shifting bits, I hand-calculated one octave and used bit shifts for the rest, then mixed the notes with PWM. **The keyboard plays 8 octaves, with chords of up to 6 clear notes.**



The complete board

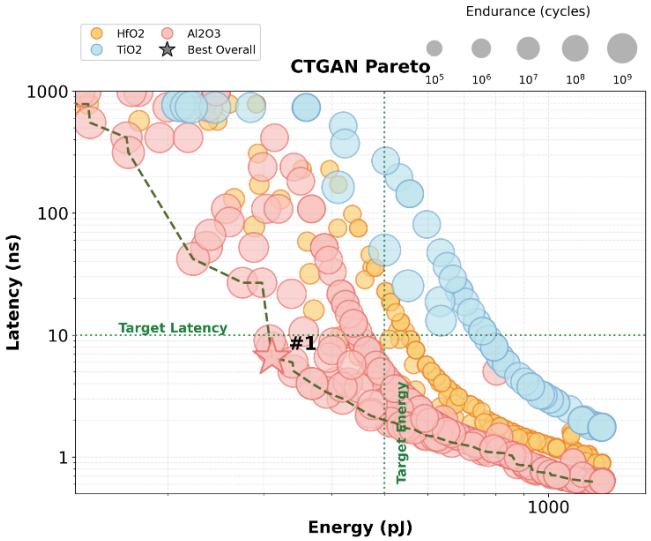


The switch that selects the octave

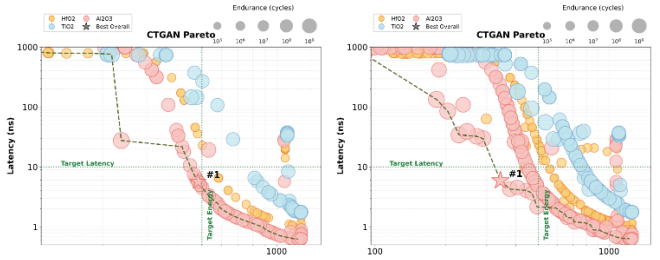
PROJECT 3 | TUFTS EMERGING CIRCUITS AND SYSTEMS LAB | SEPT. 2025 TO APRIL 2026

CTGAN Design Discovery for RRAM

The published framework I extended used a CVAE, a model that averages its training data and smooths away the extreme designs where the best performance is found. I replaced it with a CTGAN, which searches those extremes more aggressively, used the framework's physics model as a referee, and grew the training data tenfold to 5,000 samples. **The best design reached 25% higher endurance and 11% lower switching energy than the published result**, with slightly slower switching (6.57ns vs 4.07ns).



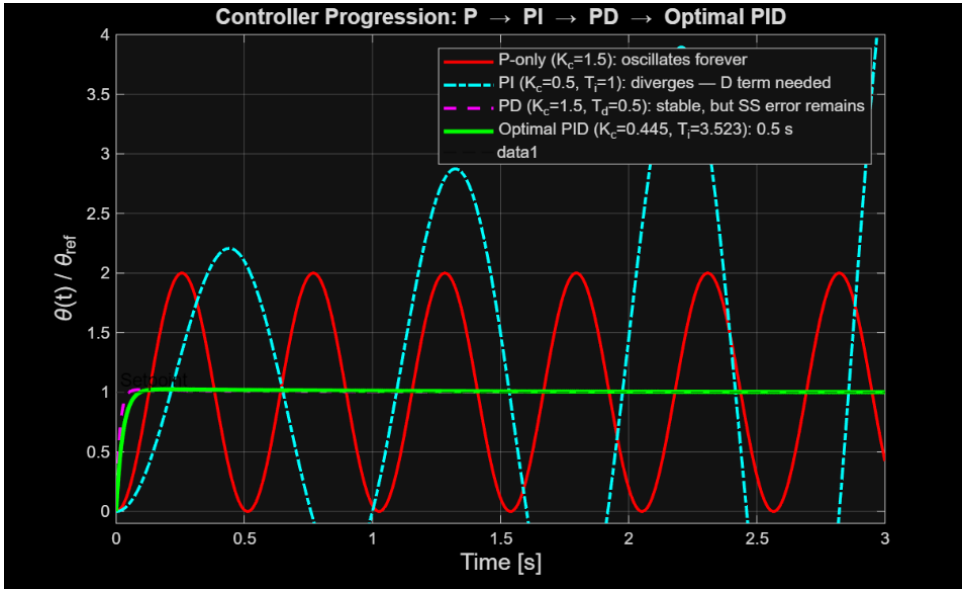
Final results with 5,000 training samples



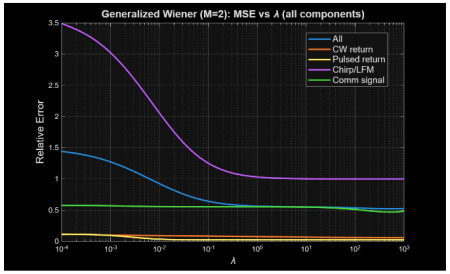
The same search with 500 samples (left) and 2,500 (right)

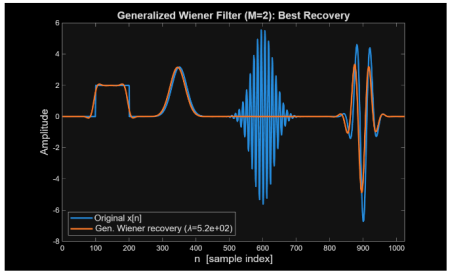
UAV Pitch Control and RF Signal Restoration

A UAV that shakes while collecting RF signals blurs them, so keeping it level comes first. Using Routh-Hurwitz analysis, I proved that P and PI controllers can never stabilize it, then searched 15,625 gain combinations for the best PID settings, reaching **a 52ms rise time and 2.6% overshoot**. I then restored a blurred, noisy RF signal with a Wiener filter, **cutting error by 77%** compared with a direct inverse.



P, PI, PD and PID controllers on the same UAV model





Wiener filter error (left) and the restored RF signal (right)